Time Series Project

In [1]:
```python
# Step 1: Read the data from the CSV file Download CSV file into a DataFrame cal
import pandas as pd
gold = pd.read_csv('gold.csv')
```

In [2]:
```python
gold.head()
```

Out[2]:

	Date	Price
0	1950-01	34.73
1	1950-02	34.73
2	1950-03	34.73
3	1950-04	34.73
4	1950-05	34.73

In [3]:
```python
# Step 2: Run the info() method, and note the data types of the two columns.
gold.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 847 entries, 0 to 846
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   Date    847 non-null    object 
 1   Price   847 non-null    float64
dtypes: float64(1), object(1)
memory usage: 13.4+ KB
```

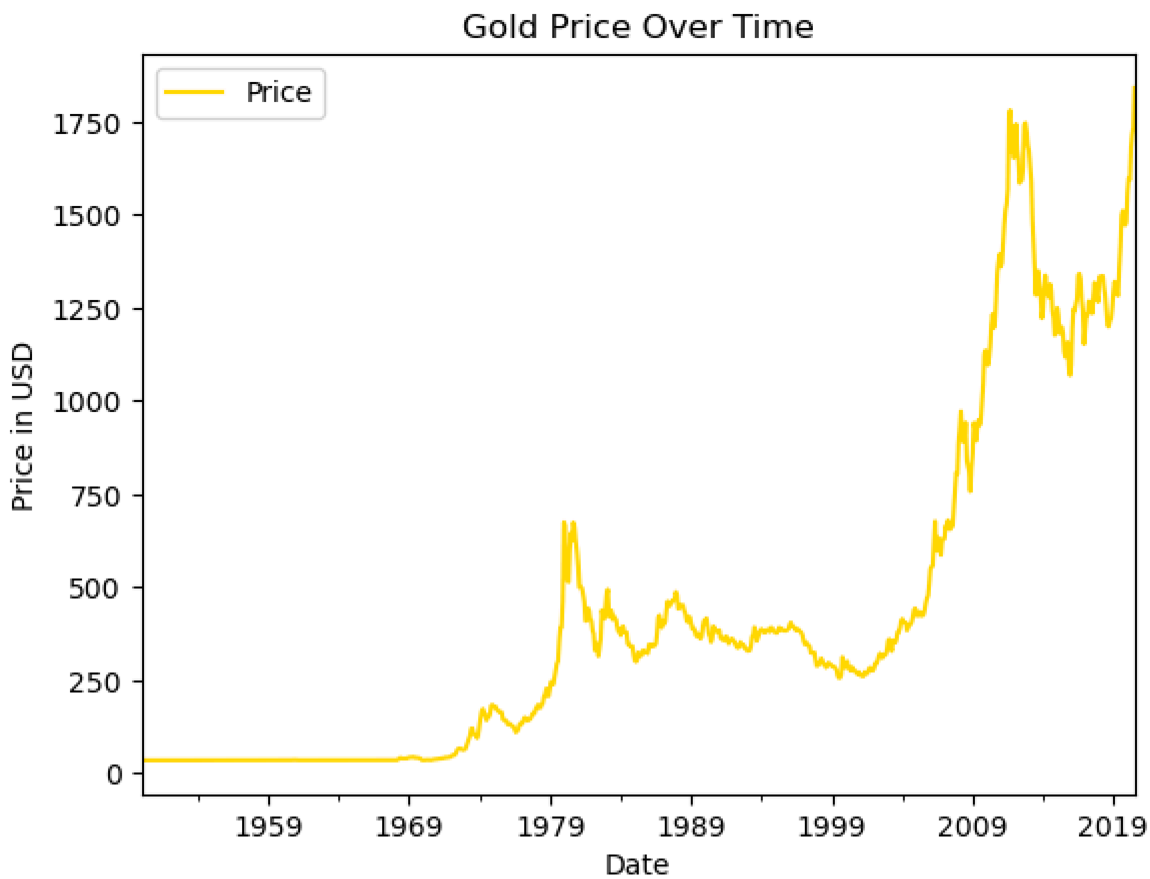
In [4]:
```python
# Step 3: Convert the Date column to the DateTime data type. Then, display the f
gold['Date'] = pd.to_datetime(gold['Date'], format = '%Y-%m')
gold.head()
```

Out[4]:

	Date	Price
0	1950-01-01	34.73
1	1950-02-01	34.73
2	1950-03-01	34.73
3	1950-04-01	34.73
4	1950-05-01	34.73

In [5]:
```python
# Step 4: Plot the data with a Pandas line plot.
gold.plot(x = 'Date', y = 'Price', color = 'gold', title = 'Gold Price Over Time
```

Out[5]:
```
<Axes: title={'center': 'Gold Price Over Time'}, xlabel='Date', ylabel='Price i
n USD'>
```



In [6]:
```
# Step 5: Index the data on the Date column. Then, generate time periods for the
gold.set_index('Date', inplace=True)
year_periods = pd.date_range(start='1950-01-01', end='2020-12-31', freq='YS')
year_periods
```

Out[6]:
```
DatetimeIndex(['1950-01-01', '1951-01-01', '1952-01-01', '1953-01-01',
               '1954-01-01', '1955-01-01', '1956-01-01', '1957-01-01',
               '1958-01-01', '1959-01-01', '1960-01-01', '1961-01-01',
               '1962-01-01', '1963-01-01', '1964-01-01', '1965-01-01',
               '1966-01-01', '1967-01-01', '1968-01-01', '1969-01-01',
               '1970-01-01', '1971-01-01', '1972-01-01', '1973-01-01',
               '1974-01-01', '1975-01-01', '1976-01-01', '1977-01-01',
               '1978-01-01', '1979-01-01', '1980-01-01', '1981-01-01',
               '1982-01-01', '1983-01-01', '1984-01-01', '1985-01-01',
               '1986-01-01', '1987-01-01', '1988-01-01', '1989-01-01',
               '1990-01-01', '1991-01-01', '1992-01-01', '1993-01-01',
               '1994-01-01', '1995-01-01', '1996-01-01', '1997-01-01',
               '1998-01-01', '1999-01-01', '2000-01-01', '2001-01-01',
               '2002-01-01', '2003-01-01', '2004-01-01', '2005-01-01',
               '2006-01-01', '2007-01-01', '2008-01-01', '2009-01-01',
               '2010-01-01', '2011-01-01', '2012-01-01', '2013-01-01',
               '2014-01-01', '2015-01-01', '2016-01-01', '2017-01-01',
               '2018-01-01', '2019-01-01', '2020-01-01'],
              dtype='datetime64[ns]', freq='AS-JAN')
```

In [7]:
```
# Step 6: Reindex the data using the new time periods, and display the first fiv
gold_reindexed = gold.reindex(year_periods)
gold_reindexed.head()
```

Out[7]:

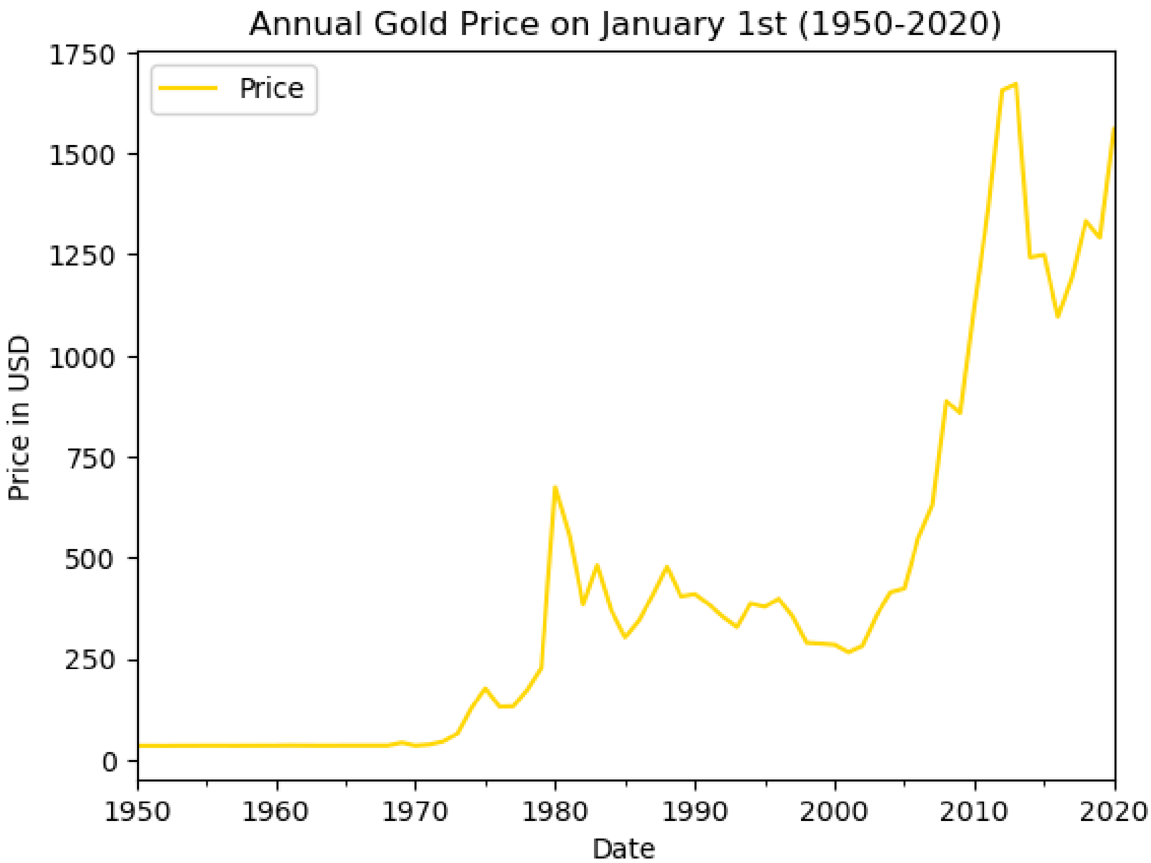
	Price
1950-01-01	34.73
1951-01-01	34.72
1952-01-01	34.49
1953-01-01	34.88
1954-01-01	34.86

In [8]:

```
# Step 7: Plot the reindexed data with a Pandas line plot.
gold_reindexed.plot(color = 'gold', title = 'Annual Gold Price on January 1st (1
```

Out[8]:

```
<Axes: title={'center': 'Annual Gold Price on January 1st (1950-2020)'}, xlabel
='Date', ylabel='Price in USD'>
```

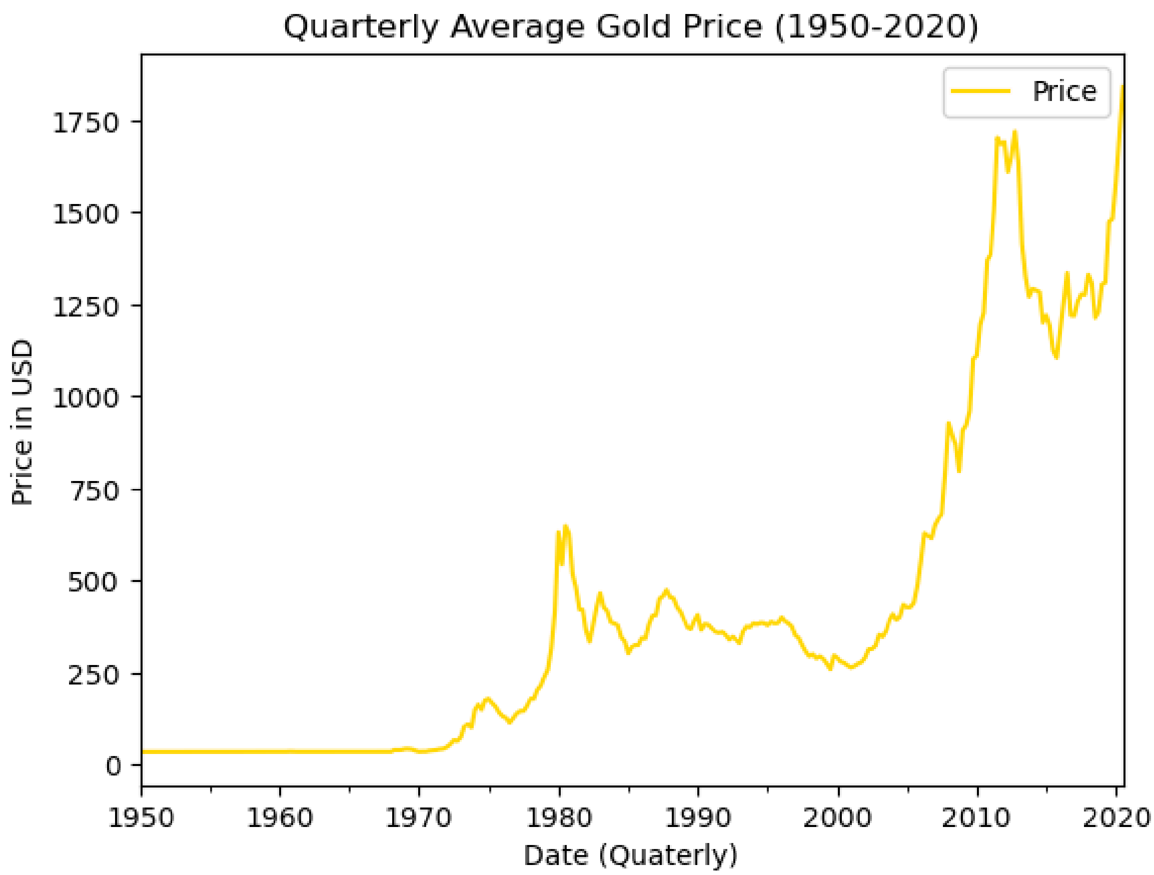


In [9]:

```
# Step 8: Resample the data to a quarterly frequency and plot the data again.
gold_quarterly = gold.resample('Q').mean()
gold_quarterly.plot(color = 'gold', title = 'Quarterly Average Gold Price (1950-
```

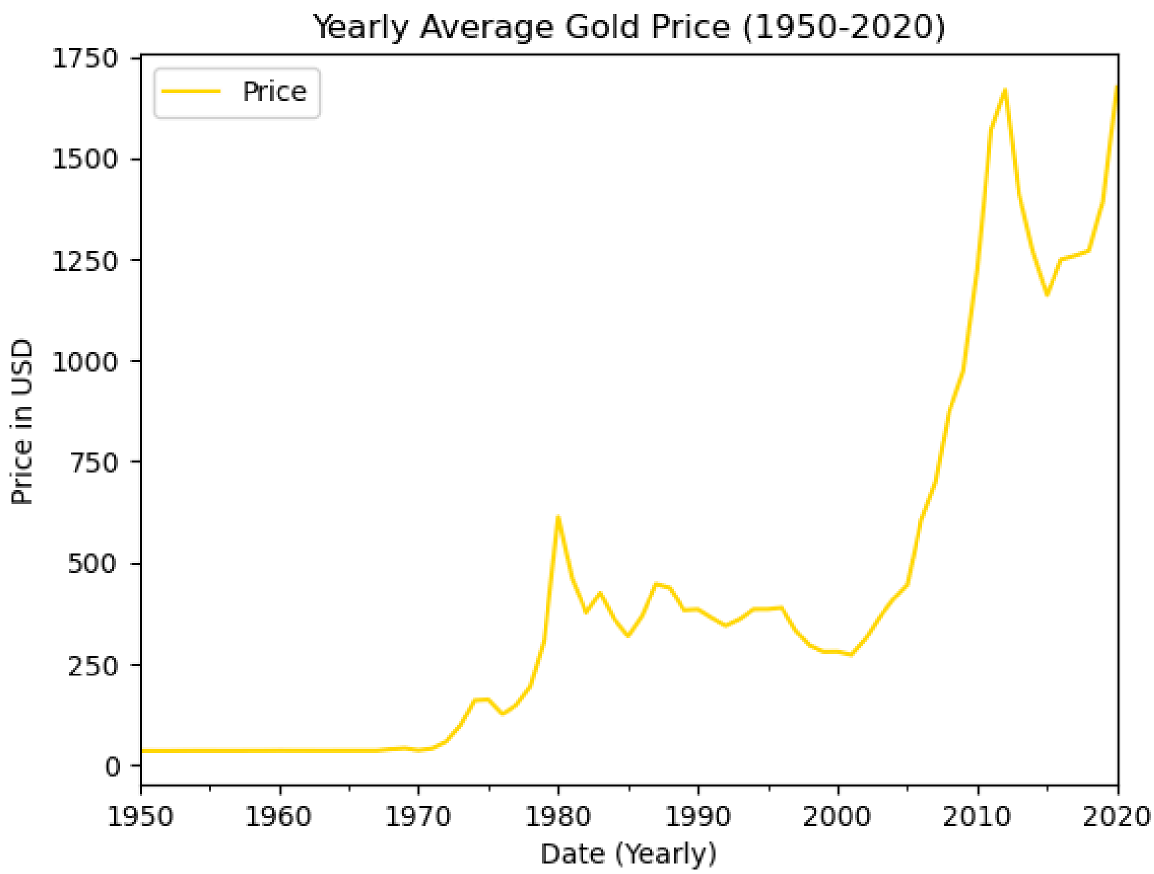
Out[9]:

```
<Axes: title={'center': 'Quarterly Average Gold Price (1950-2020)'}, xlabel='Da
te (Quaterly)', ylabel='Price in USD'>
```



In [10]:
```
# Step 9: Resample the data to a yearly frequency and plot the data one more tim
gold_yearly = gold.resample('Y').mean()
gold_yearly.plot(color = 'gold', title = 'Yearly Average Gold Price (1950-2020)'
```

Out[10]:
```
<Axes: title={'center': 'Yearly Average Gold Price (1950-2020)'}, xlabel='Date
(Yearly)', ylabel='Price in USD'>
```



In [11]:
```python
# Step 10: Plot the rolling mean for the data using a Pandas line plot. Experime
ax = gold_quarterly['Price'].rolling(window = 20).mean().plot(label = '5-Year Ro
gold_quarterly['Price'].rolling(window = 40).mean().plot(ax = ax, label = '10-Ye
gold_quarterly['Price'].rolling(window = 60).mean().plot(ax = ax, label = '15-Ye

ax.set_xlabel('Date')
ax.set_ylabel('Price in USD')
ax.legend()
```

Out[11]: <matplotlib.legend.Legend at 0x225b3f5edd0>